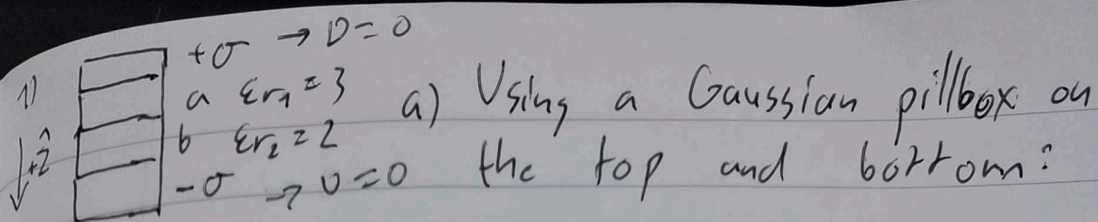
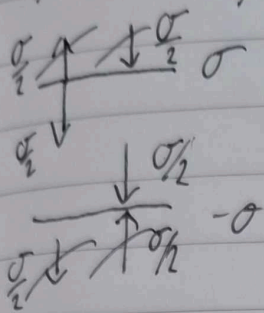


1)



$$\oint D \cdot da = q_{enc} \Rightarrow |D| \cdot 2A = \sigma A \Rightarrow |D| = \frac{\sigma}{2} \quad \text{top}$$

$$\oint D \cdot da = q_{enc} \Rightarrow |D| \cdot 2A = -\sigma A \Rightarrow |D| = \frac{-\sigma}{2} \quad \text{bottom}$$



$D = \sigma \hat{z}$ in both slabs
because they both
are experiencing
the same ^{free} charges.

$$b) \quad E = \frac{D}{\epsilon} = \frac{D}{\epsilon_r \epsilon_0} \Rightarrow \underline{E_a = \frac{\sigma}{3\epsilon_0} \hat{z}} \quad \& \quad \underline{E_b = \frac{\sigma}{2\epsilon_0} \hat{z}}$$

$$c) \quad \epsilon_r = 1 + \chi_e \Rightarrow \chi_e = \epsilon_r - 1$$

$$P = \epsilon_0 \chi_e E = \epsilon_0 E (\epsilon_r - 1) \Rightarrow \underline{P_a = \frac{2\sigma}{3} \hat{z}}$$

$$\underline{P_b = \frac{\sigma}{2} \hat{z}}$$

$$d) \quad dV = - \int_{\infty}^0 E \cdot dl = - \int_{\infty}^b 0 \, dl - \int_b^a E_b \, dl - \int_a^0 E_a \, dl$$

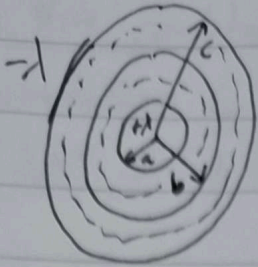
$$\Rightarrow -E_b(a-b) - E_a(0-a) = \frac{\sigma}{\epsilon_0} \left(\frac{b}{2} + \frac{a}{3} - \frac{a}{2} \right)$$

$$\Rightarrow \underline{V = \frac{\sigma}{\epsilon_0} \left(\frac{b}{2} - \frac{a}{6} \right)}$$

e) if σ is doubled, they all become doubled.

2)

2)



$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{\text{enc}}$$

$$\mathbf{D} = \epsilon_r \epsilon_0 \mathbf{E}$$

$$Q_{\text{enc}} = \int \lambda = L \lambda$$

$$\Rightarrow 2\pi s L \epsilon_r \epsilon_0 |E| = \lambda L \quad a \leq s \leq b$$

$$\Rightarrow \mathbf{E}_1 = \frac{\lambda}{2\pi s \epsilon_0} \hat{s} \quad | \epsilon_r = 1$$

$$\Rightarrow 2\pi s L \epsilon_r \epsilon_0 |E| = \lambda L \Rightarrow E_2 = \frac{\lambda}{2\pi s \epsilon} \quad b \leq s \leq c$$

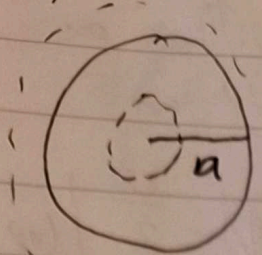
$$V = - \int_c^a \mathbf{E} \cdot d\mathbf{l} = - \frac{\lambda}{2\pi \epsilon_0} \left[\int_b^a \frac{1}{s} ds + \frac{1}{\epsilon_r} \int_c^b \frac{1}{s} ds \right]$$

$$\Rightarrow V = \frac{\lambda}{2\pi \epsilon_0} \left(\ln\left(\frac{b}{a}\right) + \frac{1}{\epsilon_r} \ln\left(\frac{c}{b}\right) \right)$$

$$C = \frac{\lambda}{V} = \frac{2\pi \epsilon_0}{\ln\left(\frac{b}{a}\right) + \frac{1}{\epsilon_r} \ln\left(\frac{c}{b}\right)}$$

3)

3)



$$r < a \quad \oint \mathbf{D} \cdot d\mathbf{a} = q_{\text{enc}}$$

$$\Rightarrow |\mathbf{D}| 4\pi r^2 = \int_0^r 4k r' \cdot 4\pi r'^2 dr' = 4\pi r^4 k$$

$$\underline{\underline{\mathbf{D} = r^2 k \hat{r}}} \quad \underline{\underline{\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0 \epsilon_r} = \frac{r^2 k \hat{r}}{\epsilon_0 \epsilon_r}}}$$

$$r > a \Rightarrow |\mathbf{D}| 4\pi r^2 = 4\pi a^4 k \Rightarrow \underline{\underline{\mathbf{D} = \frac{a^4 k}{r^2} \hat{r}}}$$

$$\underline{\underline{\mathbf{E} = \frac{a^4 k \hat{r}}{\epsilon_0 r^2}}}$$

b)

$$W_{\text{field}} = \frac{1}{2} \left[\int_{\infty}^a \frac{a^4 k^2}{\epsilon_0 r^4} \cdot 4\pi r^2 dr + \int_0^a \frac{r^4 k^2}{\epsilon_r^2 \epsilon_0} \cdot 4\pi r^2 dr \right]$$

$$\Rightarrow \frac{1}{2} \left[-\frac{a^4 k^2}{\epsilon_0} 4\pi - \frac{a^7 k^2}{7 \epsilon_r^2 \epsilon_0} 4\pi \right]$$

$$\Rightarrow \frac{2\pi a^4 k^2}{\epsilon_0} \left[-1 - \frac{1}{7 \epsilon_r^2} \right]$$

$$\underline{\underline{W_{\text{total}} = \frac{1}{2} \oint \mathbf{D} \cdot \mathbf{E} d\tau}}$$

$$\Rightarrow \frac{1}{2} \left[\int_{\infty}^a \frac{a^4 k^2}{\epsilon_0 r^4} 4\pi r^2 dr + \int_a^{\infty} \frac{r^4 k^2}{\epsilon_0 \epsilon_r} \cdot 4\pi r^2 dr \right]$$

$$\Rightarrow \frac{2\pi a^4 k^2}{\epsilon_0} \left[-1 - \frac{1}{7 \epsilon_r} \right]$$

4)

a)

North if its a normal mountain in my experience.

b)

i. A: Inward, B: Outward, C: Spinning against the clock

ii. B: Outward, B: Inward C: Spinning with the clock

c)

The left wire will have a force to the right and the right wire will have a force to the left. The wires will then repel as the magnetic field inbetween the wires cancell eachother.

5)

a)

No magnetic field is only made when a charge is moving over time.

b)

Magnetic field will be produced.

c)

i.

$$F_{mag} = Q(v \times B)$$

ii.

$$F_{mag} = Q(E + (v \times B))$$

d)

5) d) i)

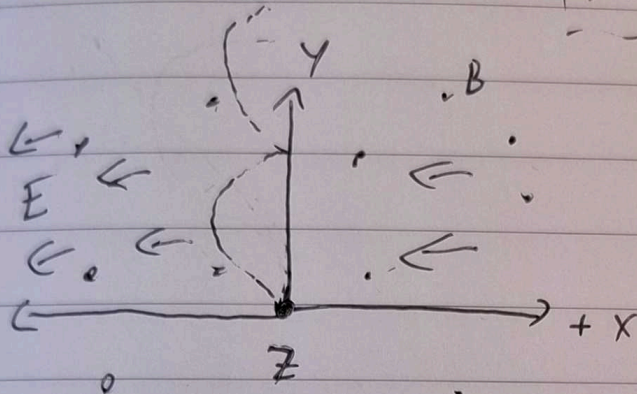
$$\vec{F}_{\text{mag}} = Q(\vec{E}(-\hat{x}) + (\frac{E}{B}\hat{y} \times B\hat{y}))$$

$$\Rightarrow -QE\hat{x} + Q(\frac{E}{B} \cdot B \cdot \sin \frac{\pi}{2} \hat{x})$$

$$\Rightarrow QE\hat{x} - QE\hat{x} = 0$$

ii) $\Rightarrow -QE\hat{x} + Q(\frac{E}{2B} \cdot B \cdot \sin \frac{\pi}{2} \hat{x})$ particle will go straight up as

$\Rightarrow Q\frac{E}{2}\hat{x} - QE\hat{x} = -Q\frac{E}{2}\hat{x}$ Magnetic and electric force cancell.



Cyclotron motion upward y and within $x < 0$.

Magnetic field does no work, so cannot add kinetic energy (velocity), but it can give an centripetal acceleration

